Question 1

Question 1: Suppose you have a binary search tree (BST) with 31 nodes.

What is the maximum possible height of the tree?
What is the minimum possible height of the tree?

Question 2: You are given a binary search tree with a height of 5.

What is the minimum number of nodes the tree can have?
What is the maximum number of nodes the tree can have?

Question 3: If a binary search tree has 63 nodes, what is the height of the tree when it is perfectly balanced?

Question 2

For the set of {1, 4, 5, 10,16,17,21} of keys, draw binary search trees of heights 2, 3, 4, 5, and 6.